

## Practice problems #11

1. Consider the matrix

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 2 \end{bmatrix}.$$

- (a) What are the eigenvalues?
  - (b) Show that you have only one linearly independent eigenvector.
  - (c) Give nullities of  $(A - 2I)$ ,  $(A - 2I)^2$ , and  $(A - 2I)^3$ , respectively.
2. Given that  $A$  is diagonalizable and

$$P^{-1}AP = D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix},$$

for some invertible matrix  $P$ , calculate the matrix exponential  $e^{At}$ .

Answers or Hints are on the next page.

#1(a): 2 (with algebraic multiplicity 3)

#1(b): What is its geometric multiplicity?

#1(c): You can find ranks of those three matrices instead?

#2:

$$e^{At} = P \begin{bmatrix} e^t & 0 & 0 \\ 0 & e^{2t} & 0 \\ 0 & 0 & e^{2t} \end{bmatrix} P^{-1}.$$

Show all steps.